

Chicken anemia virus

Karel A Schat

*Department of Microbiology and Immunology, College of Veterinary Medicine,
Cornell University, Ithaca, NY 14853, USA*

607-253-4032

607-253-3384

kas24@cornell.edu

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Abstract

Chicken anemia virus (CAV), the only member of the genus Gyrovirus of the *Circoviridae*, is an ubiquitous pathogen of chickens with a world-wide distribution. CAV shares some similarities with torque teno virus (TTV) and torque teno minivirus (TTMV) such as coding for a protein inducing apoptosis and a protein with a dual-specificity phosphatase. In contrast to TTV the genome of CAV is highly conserved. Another important difference is that CAV can be isolated in cell culture. CAV produces a single polycistronic mRNA, which is translated into 3 proteins. The promoter-enhancer region has 4 direct repeats resembling estrogen response elements. Transcription is enhanced by estrogen and repressed by at least two other transcription factors, one of which is COUP-TF1. A remarkable feature of CAV is that the virus can remain latent in gonadal tissues in the presence or absence of virus-neutralizing antibodies. In contrast to TTV, CAV can cause clinical disease and subclinical immunosuppression especially affecting CD8⁺ T lymphocytes. Clinical disease is associated with infection in newly hatched chicks lacking maternal antibodies or older chickens with a compromised humoral immune response.

9.1 Introduction

In 1979 Yuasa and co-workers (Yuasa et al. 1979) reported the presence of a new chicken pathogen causing anemia in specific-pathogen-free (SPF) chickens inoculated with herpesvirus of turkeys (HVT) to protect against Marek's disease (MD), a herpesvirus-induced T cell lymphoma. This new pathogen was first named chicken anemia agent (CAA) and is now known as chicken infectious anemia virus (CAV or CIAV). CAV is thus far the only member of the genus *Gyrovirus* which belongs to the family of *Circoviridae* together with viruses of the *Circovirus* genus (ICTVdB 2008). In addition to these two genera, the Anelloviruses, which include the human torque teno virus (TTV) and torque teno minivirus (TTMV), are often included in the *Circoviridae* (Hino and Miyata 2007). As with TTV and TTMV in humans, infection with CAV is ubiquitous in its target species (chickens) and can be found in most if not all commercial poultry operations in all continents with a poultry industry (Schat and Van Santen 2008). However, unlike TTV and TTMV, CAV has been linked to specific clinical diseases and subclinical immunosuppression although the degree of disease and immunosuppression depends on a number of factors, which will be discussed in section 9.4. This review will focus on the aspects that are relevant to the understanding of the pathogenesis and highlight some of the differences with Anelloviruses. A more detailed review of CAV has recently been published (Schat and van Santen 2008).

9.2 Historical background

The first isolation of CAV was the result of an investigation in disease problems in commercial chickens vaccinated with HVT. Yuasa et al. (1976) linked the disease problem to the presence of reticuloendotheliosis virus (REV) in HVT vaccine batches. In their subsequent study, material from one of the vaccine batches was shown to induce anemia. A highly resistant anemia-inducing agent was isolated by chick inoculation (Yuasa et al. 1979). The link to a contaminated MD vaccine suggested a link to MD vaccination as the cause of widespread distribution of CAV.

However subsequent investigations revealed that CAV was not a new pathogen associated in some way with the introduction of MDV vaccines in the early 1970's. Wellenstein (personal communication, 1989, quoted by Schat and van Santen 2008) isolated the ConnB strain of CAV from an ampoule of MD tumor cells which had been stored in liquid nitrogen at least since 1969. The tumor material was obtained from chickens experimentally infected with MD virus (MDV) which unexpectedly experienced severe anemia (Jakowski et al. 1970). This MDV strain had been passed at least 25 times in Line 7 chickens without causing anemia. Interestingly, the anemia appeared for the first time when the MDV strain was inoculated in a F2 cross of SPF x Line 7 chickens while uninfected F2 chickens remained healthy. This suggests that the SPF birds used to generate the F2 cross may have been latent carriers of CAV and that infection with MDV activated or aggravated CAV replication resulting in anemia (see section 9.4.2 for a discussion of latency and 9.4.4 for interactions between CAV and other viruses). Soiné et al. (1993, 1994) and Renshaw et al. (1996) confirmed by polymerase chain reaction (PCR) and sequencing that the ConnB strain was indeed CAV (Soiné et al. 1993). Toro et al. (2006a) confirmed that CAV infections were present in the USA as early as 1959 by analyzing banked sera from chickens used for the production of alloantisera. In Europe poultry fanciers frequently maintain old chicken breeds to show at exhibitions. These birds are not vaccinated or only vaccinated for Newcastle disease. CAV antibodies were found in 75 and 89.5% of the flocks in The Netherlands and Switzerland, respectively (Wunderwald and Hoop 2002; de Wit et al. 2004). Like TTV infections in humans, there is no indication that CAV causes any disease in these fancy breeds. The combined evidence of these studies suggests that CAV is not a new virus spread by MD vaccination, but that it is an "old" virus that has been present for a long time without causing overt problems. Based on the model proposed for the control of virus replication (see section 9.3.3.2 for details) Miller and Schat (2004) suggested that CAV has evolved as a successful pathogen that can be maintained in successive generations of chickens without causing disease. However in the modern high-stress environment of commercial poultry production it is a major pathogen that can cause significant economic problems (Schat and van Santen 2008).

9.3 Characterization of CAV and virus replication

9.3.1 Virus characteristics

9.3.1.1 Virus structure

Viruses belonging to the *Circoviridae* share a number of characteristics but there are also major differences between the different genera (see also chapter 6). All viruses belonging to this family have a small covalently closed circular single stranded DNA genome. CAV and Anelloviruses both have a negative sense genome in contrast to viruses in the genus *Circovirus*. The genome of CAV consists of 2298 or 2319 nucleotides (nt) (Claessens et al. 1991; Noteborn et al. 1991). The difference between these two sequences is the addition of an extra direct repeat (DR) of 21 nt in the promoter enhancer region (see section 9.3.3.2). Limited number of passages (n=20 to 30) in the MDV-derived chicken cell line (MDCC)-MSB1 was reported to result in the insertion of the 5th DR without influencing the pathogenicity (Todd et al. 1995). However other researchers, using a different isolate, were unable to demonstrate the insertion of a 5th DR after 129 passages (Chowdhury et al. 2003).

9.3.1.2 Resistance to physical and chemical treatments

Other shared characteristics of the *Circoviridae* are the small size of the virus particles and the extreme resistance to physical and chemical treatments. This combination complicates the elimination of these viruses from other pathogens especially if cell culture passage is not possible or not desirable. CAV has an average size of 25 nm (Gelderblom et al. 1989; Crowther et al. 2003) while porcine circovirus type 2 (PCV-2), an important pathogen in pigs, has an average size of 20.5 nm (Crowther et al. 2003). CAV and PCV-2 are highly resistant to many common disinfectants and heat treatment (Yuasa et al. 1979; Yuasa 1992; Urlings et al. 1993; Opriessnig et al. 2007). The resistance of *Circoviridae* is a concern for the preparation of blood products for human use especially in view of the widespread occurrence of TTV and TTMV. Commonly used procedures for virus inactivation in plasma products are based on the use of detergents, pasteurization of products in

solution or dry heating of lyophilized products. Welch et al. (2006) used CAV and PCV-2 as a substitute for TTV to determine the effectiveness of virus inactivation by pasteurization and dry heat. Pasteurization of CAV in heat-treated human FVIII at 60°C for 24 hours reduced the titer by 1.42 log, but increasing the temperature to 75°C reduced the titer > 3.5 log after 30 minutes. Heating of lyophilized material at 80°C for 72 hours or 120°C for 30 minutes reduced the virus titers only 10-fold. Inactivation of CAV in minced chicken meat required >10 minutes at 90°C (Urlings et al. 1993).

9.3.1.3 Genetic variation

Thus far most CAV isolates belong to the same serotype although a second serotype has been reported (Spackman et al. 2002a; 2002b) based on physicochemical characteristics and pathology. Interestingly, the second serotype was isolated using MDCC-CU147, which is approximately 10-100 times more susceptible to infection with serotype 1 strains than the commonly used MDCC-MSB1 cell line (Calnek et al. 2000) (see section 9.3.2). There was no cross neutralization between the two serotypes and a commercial ELISA test did not detect antibodies to the serotype 2 virus. CAV-specific primers failed to amplify serotype 2 DNA. Based on the lack of antibody detection in the ELISA, the negative PCR results and absence of information on the genome it is not certain that this pathogen is even a circovirus. Many of the reported physicochemical characteristics are similar to parvoviruses and picornaviruses. As a consequence all strains are considered to belong to the same serotype in this review.

Fig. 9.1

A major difference between CAV and the Anelloviruses is the genetic stability of the former. Isolates separated by time and geographical location show very little variation at the nucleotide level or the amino acid level for viral protein 1 (VP1). For example, Renshaw et al. (1996) reported that the predicted amino acid sequence for VP1 of the ConnB isolate (U69548) obtained from material frozen in 1969 (see section 9.2) has only 5 or 6 amino acid differences with the Cux-1 isolate based on the sequence provided by Noteborn et al. (1991) (M55918) and Meehan et al. (1992)

(M81223), respectively. Cux-1 was isolated in Germany (von Bülow et al. 1983) while the ConnB isolate came from Connecticut in the USA. Recently a large number of isolates have been fully or partially sequenced. These isolates were obtained in Bangladesh (Islam et al. 2002), Brazil (Nogueira et al. 2007), China (He et al. 2007; Ducatez et al. 2008), Malaysia (Chowdhury et al. 2003), Nigeria (Ducatez et al. 2006), Slovenia (Krapež et al. 2006) and the USA (van Santen et al. 2001; van Santen et al. 2007). Comparison of all sequence data indicates that these isolates can be divided into 3 or 4 distinct groups, with “old” and “new” isolates scattered throughout the different groups. The overall number of nucleotide differences for the VP1 sequence is limited and most reports indicate that the maximum difference between strains at the amino acid level is less than 6% (e.g., Ducatez et al. 2006; 2008, see Fig 9.1). The differences between the predicted amino acid sequences for VP2 and VP1 are even smaller and the promoter/enhancer region has very few differences. He et al. (2007) reported that recombination between genotypes can occur, which may lead to new genotypes. Thus far the minor differences between genotypes have not been linked to clear differences in pathogenicity (reviewed by Schat and van Santen 2008).

The lack of genetic diversity of CAV contrasts sharply with the high degree of genetic diversity described for TTV and TTMV. Ducatez et al (2008) suggested that VP1 is under strong negative selection especially in countries where chickens are not vaccinated. However, humans are not vaccinated against TTV or TTMV and vaccination-driven diversity for these two virus families seems therefore unlikely. A second explanation offered by these authors is the exquisite relationship between virus and host, which may lead to vertical transmission of virus or viral DNA as proposed by Miller and Schat (2004) (see sections 9.3.3.2 and 9.4.2).

9.3.2 Virus isolation and propagation

Thus far, CAV, PCV-1 and PCV-2 are the only members of the *Circoviridae* which can be readily isolated in cell culture. Desai et al. (2005) reported that TTV can be propagated in the Chang liver cell line and PHA-stimulated peripheral blood mononuclear cells. Hino and Miyata (2007) questioned the efficacy of virus replication in these cultures.

Yuasa (1983) propagated CAV in MDCC-MSB1 and -JP 2 and the avian leukosis virus -transformed B cell line LSCC-1104B1, although MDCC-RP1 and -BP1 and LSCC-1104X5 and -TLT were refractory to infection. Not all CAV strains could be isolated in MDCC-MSB1 (Soiné et al. 1994; Renshaw et al. 1996; Islam et al. 2002; Nogueira et al. 2007; van Santen et al. 2007). However, one of these isolates, CIA-1 (Lucio et al. 1990), could be propagated in the MSB1(S) subline albeit to lower titers than Cux-1 (Renshaw et al. 1996). The difference in infectivity for MSB1 cells between CIA-1 and Cux-1 was originally thought to be related to a 316 bp fragment which included the hypervariable region of VP1 with perhaps secondary influences by differences in VP2 and VP3 (Renshaw et al. 1996). Subsequent examination of additional isolates that could not be propagated in MSB1 cells failed to reveal a consensus sequence responsible for the lack of propagation.

Calnek et al. (2000) examined 26 MDCC lines representing $CD4^+CD8^-$, $CD4^-CD8^+$ and $CD4^-CD8^-$ cells expressing either T cell receptors (TCR) $\alpha\beta1$ or TCR $\alpha\beta2$. Several cell lines representing the different phenotypes could be infected, but MDCC-CU147 ($CD4^-CD8^+$) was significantly more susceptible than MSB1 (S) for the propagation of Cux-1. This cell line was approximately 100 to 1000-fold more susceptible to infection with CIA-1 than MSB1 (S). Recently van Santen et al. (2007) confirmed that CU147 can be used successfully for the propagation of CAV isolates that could not be isolated in their MSB1 subline. The reason(s) for the differences in susceptibility between cell lines have not been elucidated.

9.3.3 Virus replication

9.3.3.1 DNA replication and viral transcription

It is assumed that CAV enters cells through binding to cell surface receptors but specific receptors have not been identified. The finding that different subsets of T cells as well as specific ALV-transformed B cell lines can be infected (see 9.3.2) does not clarify this question. Claessens et al. (1991) identified a prominent hairpin structure in the sequence, which was later identified as a nuclease S1 site (Todd et al. 1996; Noteborn et al. 1992). CAV, lacking the machinery to reproduce its own DNA, depends on dividing host cells for the replication of DNA, which requires the

formation of double-stranded replicative form (dsRF). The actual mechanism of DNA replication has not been resolved. Todd et al. (1996; 2001) were unable to conclusively determine if replication occurred through the rolling circle replication (RCR) model proposed for PCV-1 and PCV-2. Arguments against the RCR model for CAV are the absence of a replicase-associated (Rep) protein and the absence of sequences in VP1 needed for the binding of dNTP. In addition the conserved stem-loop structure with the nt sequence TAGTATTAC, which is associated with the initiation of the RCR-based replication of DNA in PCV and geminiviruses (Bassami et al. 1998), is only semi-conserved in CAV (nt 42-50 using Accession number M81223, Meehan et al. 1992). On the other hand, three peptide motifs typically associated with RCR DNA replication are present in CAV VP1.

Fig. 9.2

Once the virus or viral DNA has entered the cells viral transcripts can be detected as early as eight hours post infection (pi). Most authors reported the presence of a single, polycistronic message of approximately 2.1 kb (Noteborn et al. 1992; Phenix et al. 1994). Using Northern blot analysis Kamada et al. (2006) found two additional polyadenylated transcripts of 1.6, and 1.3 to 1.2 kb. These RNA species appeared between 48 to 72 hours pi in infected but not control MSB1 cells. All transcripts were amplified by RT-PCR; three different clones were obtained in addition to a clone representing the full length transcript. The 1.3 kb clone used a splice donor and acceptor site at nt 1222 and 1814, while the smaller one of 1.2 kb used a splice donor and acceptor site at nt 994 and 1095 in addition to the ones used by the larger RNA species. Finally, the 0.8 kb clone may represent an artifact because the 5' and 3' flanking sequences are identical suggesting the possibility of homologous recombination. Excluding the potential proteins from the 0.8 kb clone, the 1.3 and 1.2 kb clone could account for two additional VP1 and one additional VP2 based proteins (Fig. 9.2). Thus far, proteins associated with these transcripts have not been demonstrated and the relevance of these new findings for the biology of CAV is poorly understood at the present time. It will be of considerable interest to determine the fate of virus replication *in vitro* and *in vivo* when the different splice donor sites are modified and the new transcripts are no longer expressed.

9.3.3.2 Transcriptional control

Fig. 9.3

The 5' nontranscribed region of the CAV genome has been identified as the sole promoter/enhancer region controlling viral transcription and replication. This region contains four or five DR with a 12 bp insert after the first two DRs and several known enhancer-like elements (Fig. 9.3A) (Noteborn et al. 1991). The first two DR and 12 bp insert are the dominant elements based on experiments in which different deletion constructs were used to drive the expression of Cm acetyltransferase. The region encompassing the DR can enhance the thymidine kinase promoter independent of its orientation when inserted upstream of the promoter. The DR has a consensus sequence for the cyclic AMP response element binding protein, but it binds an unidentified T cell nuclear protein instead. The 12 bp insert binds the SP1 protein (Noteborn. et al. 1994b). Virus constructs lacking the complete DR region or containing three repeats but lacking the 12 bp insert were not viable, while additions of 6- to 24 nt to the 12 bp insert reduced virus replication significantly (Noteborn et al. 1998b).

Cardona et al. (2000b) reported that ovaries of antibody-positive and – negative SPF hens can be positive for virus DNA (see section 9.4.2). This finding in combination with observations from the SPF poultry industry that birds often seroconvert when in production (Schat and van Santen 2008), led Miller et al. (2005) to hypothesize that CAV expression could be influenced by the hormonal regulation of the reproductive system. Inspection of the DR region suggested that the ACGTCA sequence in the repeats separated by 15 nt could function as a hormone response element (HRE). An optimal HRE typically consists of two inverted repeats separated by 15 nt, but direct repeats separated by 15 nt can also function as a HRE especially if SP1 or NFY binding sites are located in close proximity (Wang et al. 1999; Saville et al. 2000; Safe 2001). To test the hypothesis that the CAV promoter/enhancer activity could be regulated by hormonal influences Miller et al. (2000) generated two short (SE and S5) and two long (LE and L5) promoter constructs (Fig. 9.3A) to drive expression of enhanced green fluorescent protein (EGFP). The differences between SE and LE *versus* S5 and L5 are that the former contains some potential start and stop

codons from the cloning vector, which could affect translation. Transfection of the chicken liver cell line LMH and the LMH-2A, which overexpresses the estrogen receptor (Sensel et al. 1994), showed a significantly enhanced expression in the presence of estrogen (Fig. 9.3B) suggesting that the HRE can function as an estrogen response element (ERE). It was also noted that the long promoter constructs induced a lower EGFP expression than the short forms. Real-time RT-PCR experiments indicated that the decreased response was at the transcriptional level (Fig. 9.3C). The decreased response was linked to an E-box-like sequence at the transcription start point (Miller, Jarosinski and Schat, manuscript in preparation). In addition to the upregulation of expression by estrogen, Miller, Jarosinski and Schat (manuscript in preparation) also found that the chicken ovalbumin upstream promoter transcription factor 1 (COUP-TF1) can negatively regulate transcription of EGFP by binding to the short promoter form. Clearly, regulation of expression of CAV is complex and may be dose-dependent with low virus levels being controlled by a balance between COUP-TF1 and E-box-binding proteins downregulating and estrogen upregulating virus replication.

9.3.3.3 Functions of viral proteins

As predicted by the initial sequence information (Claessens et al. 1991; Noteborn et al. 1991; Meehan et al. 1992) three viral proteins, VP1, VP2 and VP3, have been identified in virus infected cells (Chandratilleke et al. 1991). These proteins are translated from a single polycistronic 2.1 kb mRNA by alternate use of initiation codons. VP1 is using the fifth AUG as the start codon (Noteborn et al. 1992), but the actual mechanism of alternate codon use has not been elucidated. Douglas et al. (1995) reported that VP2 and VP3 could be detected in infected cells as early as 12 hours pi, while VP1 was not present until 30 hours.

The 52 kDa VP1 is the only structural protein forming the viral capsid (Todd et al. 1990). The N-terminal region has limited similarity to protamines (Claessens et al. 1991) suggesting a DNA-binding function (Todd et al. 2001). There is a lack of information on the importance of the many differences in amino acid sequence for the tertiary structure and antigenicity of VP1. Thus far polyclonal chicken anti-CAV sera recognize all strains that have been tested, but only one neutralizing epitope has been

mapped using monoclonal antibody (MAb) 2A9 (McNulty et al. 1990; Scott et al. 2001). The formation of epitopes recognized by the polyclonal virus neutralizing (VN) chicken antibodies requires the expression of VP1 and VP2 in the same cells (Koch et al. 1995; Noteborn et al. 1998a) suggesting that VP2 acts as a scaffold protein. Epitope mapping by pepscan analysis of CAV proteins using a panel of MAb has only been reported for VP2. Most MAb reacted with epitopes located between amino acids 111-126 and one MAb reacted with an epitope between amino acids 21-36 (Wang et al. 2007).

Peters et al. (2002; 2005) reported a second function for the 24 kDa VP2. It has a dual-specificity phosphatase (DSP) activity catalyzing the removal of phosphatase from phosphoserine/phosphothreonine and phosphotyrosine substrates. The protein coded by ORF2 of TTMV also has DSP activity and the signature sequence for DSP activity is also present in TTV (Peters et al. 2002). The signature motif for the DSP activity in VP2 includes the amino acids ICNCGQFRKH at position 94-103. The underlined amino acids are conserved in the DSP protein family. The mutation C95S completely removed all DSP activity, while the mutation C97S reduced the phosphotyrosine activity by 30% but drastically increased the phosphoserine/phosphothreonine activity. The double mutant had no activity with either substrate. Virus titers were reduced significantly in MSB1 cells infected with the C95S and the C97S mutants. The functional importance of the DSP activity is still unknown and awaits identification of the substrate.

Fig. 9.4

A key pathological finding in CAV-infected chickens is the transient depletion of thymocytes in the cortex (reviewed by Schat and van Santen 2008) caused by apoptosis (Jeurissen et al. 1992b). The apoptosis is induced by the 13.6 KDa VP3 although VP2 may also have some apoptotic activity (Noteborn. 2004). Truncation of VP3 by removing amino acids 110-120 significantly reduces the apoptosis in MDCC-MSB1 cells (Noteborn et al. 1994a), but the actual mechanism of apoptosis in non-transformed chicken cells has not been elucidated. It is of interest that the cortex of virus-infected chickens can be almost completely depleted while *in situ* hybridization or immunohistochemistry show only a relatively small number of infected cells (Fig.

9.4.) (Allan et al. 1993; Hu et al. 1993b). Perhaps VP3 can be “recycled” into new cells, or the VP3-induced apoptosis causes a “bystander effect” affecting non-infected cells or the *in situ*-hybridization severely underestimated the number of virus-infected cells. The finding that VP3 or Apoptin® selectively induces apoptosis in human cancer cells has generated considerable interest. An apoptosis-inducing protein, dubbed TAIP, has also been identified in TTV (Kooistra et al. 2004). The mechanisms of Apoptin® and TAIP-induced apoptosis in human cells are discussed in chapter 10.

9.3.4 Attenuation

Von Bülow and Fuchs (1986) demonstrated that Cux-1 could be attenuated by serial passage in MDCC-MSB1. Attenuation was noted after passage (p) 49 but residual pathogenicity was still present after 100 passages. Todd and collaborators (1995) confirmed that Cux-1 could be attenuated by passage in MDCC-MSB1, but even at p173 some residual pathogenicity was noted. Molecularly cloned viruses were obtained from p173, some of which were attenuated (e.g., clone 10) while other clones were still fully pathogenic (e.g., clone 15). Unfortunately clone 10 reverted to pathogenic virus after ten passages in one-day-old chicks. The revertant virus contained several minor subpopulations with different mutations and it was suggested that some of the minor subpopulations could have an increased pathogenicity (Todd et al. 2003). Subsequently, Cux 1 was examined for pathogenicity and antigenicity at p310 using pooled material and clones indicating that there was still residual pathogenicity (Scott et al. 1999). Sequence analysis of VP1 indicated that 6 amino acids were changed in most of the clones compared to p13 or the original Cux-1 sequence. The relevance of these changes for the pathogenicity is not clear because clone 33 was more pathogenic than p13, while clones 31 and 34 were highly attenuated. These three clones differ only at amino acid position 89 where clone 33 has a T, like the p13 and p173 clone 10, while clones 31 and 34 have an A. Clone 33 is neutralized by MAb 2A9 which recognizes a conformational epitope. The sequence of the population that is not neutralized by MAb 2A9 is similar to clone 34. (Scott et al. 2001). Using chimeras, Todd et al (2002) showed that amino acid 89 was responsible for the differences between clones 33 and 34 in pathogenicity and

antigenicity. Site-directed mutagenesis of this amino acid from T to A did not change the pathogenicity of the cloned, low passage Cux-1 strain. Only one other change was reported in VP3 where at position 41 clone 33 has an I, while the other two clones and Cux-1 have a T (Scott et al. 2001), but this change was not responsible for the difference in pathogenicity (Todd et al. 2002). Interestingly the three clones have a G at position 186 of VP2 instead of an E, which was also a change made by Peters et al. (2006) by site-directed mutagenesis (see below).

Table 9.1

A low-pathogenic virus was obtained by analyzing different clones from p12 of the pathogenic AH9410 isolate (Yamaguchi et al. 2001). Only one change in VP1 (Q394H) was associated with a decrease in pathogenicity, but there were still residual lesions with the clones expressing H at amino acid position 394 such as decreased hematocrit values. Site-directed mutagenesis changing the H > Q and *vice versa* changed the lesion score as expected. Inspection of differences between Cux-1, its cloned apathogenic derivative C 34 (Scott et al. 2001) and the high (AH-C364) and low (AQH-C140) pathogenic clones (Yamaguchi et al. 2001) indicates that none of the changes in VP1 can be linked directly to pathogenicity (Table 1). A possible explanation is that different individual changes may affect the tertiary structure of VP1 causing subtle changes on the virion surface and hence impair the interactions with the putative virus receptor.

Table 9.2

Peters et al. (2006) introduced specific mutations in VP2 to analyze the possibility of attenuating CAV. Mutants 1-4 were generated in the catalytic motif of DSP activity and 5-11 in the region expected to have a high degree of secondary structure (Table 2). These mutants were evaluated in cell culture (Peters et al. 2006) and by inoculation in 7-day-old embryos (Peters et al. 2007) and 1-day-old chickens (Kaffashi et al. 2008). Some of their findings are summarized in Table 9.2. With the exception of mutant 3 all could be propagated in MDCC-MSB1, causing relative differences in the degree of cytopathic effects. Mutant 3 did not propagate to levels allowing experimental studies. Mutants 5, 6, 7 and 9 did not cause apoptosis although VP3 could be detected in the cytoplasm supporting the suggestion by Noteborn

(2004) that VP2 may play a role in the induction of apoptosis. Unfortunately data were not provided for the viruses with mutations in the DSP signature sequence. Interestingly, the wild type CAV caused downregulation of major histocompatibility complex (MHC) class I on the surface of MDCC-MSB1 cells. To determine if the mutants were attenuated 7-day-old embryos were inoculated in the yolk sac and examined for lesions in the lymphoid organs and for hemorrhages. All mutants caused significantly fewer lesions than CAU269/7, but the lesion score of mutant 11 was the only one that was not significantly different from the uninfected control (Peters et al. 2007). Infection of one-day-old birds indicated that the relative thymus weight was decreased at 14 days pi for mutant 7 and 11, but not for mutant 6. In contrast, the thymic cortex was normal for mutant 7 and 11. A spontaneous new mutant (S77N) was identified in cell cultures infected with mutant 5 (R129G) (Kaffashi et al. 2008) raising some interesting questions about the long-term stability of the VP2 mutants. The mutation E186G (mutant 11) was also detected in the high passage pathogenic and apathogenic clones 33 and 34 described by Scott et al. (2001).

It is clear from these data that attenuation of CAV is complex and that many individual changes may lead to partial or complete attenuation. The approach of site-directed mutagenesis is certainly an important avenue to clarify the role of individual amino acids in the pathogenicity of CAV, but it may be necessary to introduce several changes in VP1 and VP2 and perhaps VP3 to obtain a truly attenuated stable virus that can be used as a vaccine.

9.4 Pathogenesis

Infection with CAV can have different consequences depending on whether the birds are kept as commercial production flocks, referred to as conventional chickens in this chapter, or as SPF flocks. The pathogenesis of infection and disease in the former is fairly well understood and is reviewed in section 9.4.1. Infection with CAV is not uncommon in research or commercial SPF flocks but the maintenance of infection in these flocks is poorly understood and is discussed in section 9.4.2. The consequences of infection for the immune response and interactions with other pathogens are detailed in sections 9.4.3 and 9.4.4, respectively.

9.4.1 Pathogenesis in conventional chickens

The consequences of infection depend on the combination of age of infection and the presence of maternal antibodies (reviewed by Schat and van Santen 2008). Infection of young chickens, in general less than 2 weeks of age, may result in clinical disease and mortality in the absence of maternal antibodies (Yuasa et al. 1980a). Likewise clinical infections may occur in older birds when antibody responses are impaired for example, by infectious bursal disease virus (IBDV) (Yuasa et al. 1980b) or by embryonal bursectomy (Hu et al. 1993a). In general infection cannot be established in young chickens with maternal antibodies against CAV (Yuasa et al. 1980a), which can be protective against infection until 3 or even 4 weeks of age when antibodies are not longer detectable by commercial Elisa kits (Markowski-Grimsrud and Schat 2003). Infections occur frequently in chickens once maternal antibodies wane, but infections are subclinical unless the antibody response has been damaged by infections with immunosuppressive viruses such as IBDV or MDV.

9.4.1.1 Infection in young chickens without maternal antibodies

Chicks hatched from eggs inoculated in the yolk-sac as 6-day-old embryos from antibody-negative hens or from eggs produced from experimentally infected, antibody-negative hens developed clinical disease (Yuasa and Yoshida 1983; Hoop 1992). Outbreaks of disease in commercial flocks have indeed been linked to late or uneven seroconversion just prior to or during egg production (Chettle et al. 1989; Engström 1999; Davidson et al. 2004) resulting in vertical transmission. Insemination with semen from viremic males may also result in vertical transmission (Hoop 1993). Testing for antibodies around 10 to 12 weeks of age followed by vaccination if exposure of the breeding flock has been insufficient will solve the problem the vertical transmission (Smith 2006).

In addition to vertical transmission chickens can also become infected by horizontal infection presumably by the fecal-oral route, which is the most likely route of exposure in flocks hatched with maternal antibodies (Yuasa et al. 1983). Virus was isolated up to 49 days of age from rectal contents when chickens were inoculated at one-day of age, even while VN antibodies were detected at 21 days pi. Another

potential source of infection is transmission from feathers. Virus was detected between 7 and 14 days pi in feathers from birds infected with 3 different CAV isolates. Feather-extracted CAV was used to successfully infect newly hatched chickens by dripping the extract on the eye and oral mucosal surfaces (Davidson et al. 2008).

The pathogenesis of infection in one-day-old chickens has been studied in detail by experimentally infecting chickens by injection (Taniguchi et al. 1983; Yuasa et al. 1983; Hu et al. 1993b; Smyth et al. 1993; van Santen et al. 2004), or orally (van Santen et al. 2004; Kaffashi et al. 2006). In general, high virus doses of 300 to 3000 chick infective doses 50% (CID₅₀) or over 10⁴ tissue culture infective doses 50% (TCID₅₀) were used, which may represent actual exposure levels in commercial poultry operations. Virus could be isolated at low titers from many tissues as early as one day pi but titers increase rapidly afterwards (Yuasa et al. 1983). Within three to four days pi viral antigen or viral genomes can be detected initially in the lymphoid organs and bone marrow and subsequently in many other organs including the intestinal tract and skin. Viral antigens were mostly associated with lymphoid aggregates (Smyth et al. 1993). Virus shedding into the intestinal tract most likely occurs via infected lymphocytes with perhaps the cecal tonsils as an important source (van Santen et al. 2004) or perhaps through urine, as kidneys were consistently positive between six and 14 days pi. The finding of an occasional bird with viral antigen in the feather pulp (Smyth et al. 1993) suggested the possibility of virus shedding through feathers, which was later confirmed by Davidson et al. (2008). Once VN antibodies develop viral antigens disappear. Interestingly virus could be reisolated from peripheral blood cells but not from serum for a long period after VN antibodies were present (Imai et al, unpublished data quoted by Yuasa 1994). This observation suggests that CAV can persist as a latent virus, a concept that was first proposed by McNulty (1991) and subsequently by Miller and Schat (2004) (see section 9.4.2 for details). Recently, studies on the pathogenesis have used real-time quantitative (qPCR) assays (Kaffashi et al. 2006), but the results of these studies have not changed our fundamental understanding of the pathogenesis. Unfortunately, with one exception (Markowski-Grimsrud and Schat 2003), these studies used only qPCR,

which does not provide information on the actual timing and intensity of virus replication based on the quantitative analysis of transcripts by qRT-PCR.

The target cells for CAV replication are dividing cells of the hemocytoblast and T cell lineages (reviewed by Adair 2000; Miller and Schat 2004).

Hemocytoblasts are the precursors for erythrocytes, thrombocytes, and heterophils (which are the mammalian equivalent of neutrophils) and infection with CAV causes a temporal decrease in these cells (Taniguchi et al. 1983). The infection in the hemocytoblasts is responsible for clinical signs such as anemia, which is often defined as hematocrit values below 27%, hemorrhages and secondary bacterial infections due to a decrease in innate immune responses. The decrease in neutrophils and thrombocytes, which have a major role in innate immune responses, contribute to the increased susceptibility to bacterial disease often leading to dermatitis or “blue-wing” disease (Engström and Luthman 1984). Infection of the T cell precursors expressing CD3 in the cortex of the thymus causes a decrease in T lymphocytes expressing CD4 and CD8 (Adair et al. 1993; Hu et al. 1993b). Infection also occurs in spleen lymphocytes especially in CD8⁺ cells expressing TCRαβ (Adair et al. 1993). Cytotoxic T lymphocytes (CTL) are expressing CD8 and TCRαβ, while natural killer (NK) cells express CD8 but not TCRαβ. These findings suggest that CAV infection may impair developing T cell-mediated immune responses. Markowski-Grimsrud and Schat (2003) found a significant decrease in CTL responses to REV 7 days pi with REV and CAV. Replication of CAV was demonstrated by qRT-PCR and q PCR assays at that time, presumably leading to apoptosis of the developing REV-specific CTL.

The consequences of the infection are damage to the bone marrow cells with a replacement of the normal structure by fat cells and thymus atrophy with mostly a collapse of the cortex. These changes are often reversible unless the birds die as a consequence of anemia or secondary bacterial infections. Significant loss of body weight may be noticed under experimental conditions (e.g., Hu et al. 1993b; Kaffashi et al. 2006).

9.4.1.2 Infection in immunocompetent chickens

Infection of immunocompetent chickens is often subclinical but Smyth et al. (2006) found that experimental challenge of three- and six-week-old chickens resulted in damage of the thymus cortex. Interestingly, very few antigen-positive cells could be detected in the bone marrow and none of the chickens developed anemia or other clinical signs. Infection of six-week-old birds follows the same viremia pattern in thymus, liver and spleen as infection at one day of age but lesions did not develop when birds were infected at 6 weeks of age (Kaffashi et al. 2006). However, CAV infection caused anemia and other lesions when the embryonally bursectomized birds were infected at five weeks of age (Hu et al. 1993a). Likewise, infection with very virulent IBDV can prolong the persistence of replicating CAV (Imai et al. 1999), which may lead to CAV-associated pathology. These experiments clearly indicate that age resistance is related to the ability to develop VN antibodies in a timely fashion and that it is not related to the disappearance of specific developmental stages of T lymphocytes as originally proposed by Jeurissen et al. (1992a).

9.4.1.3 Factors influencing the pathogenesis

Genetic differences between birds, virus strains and route of infection are possible factors that may influence the pathogenesis in addition to the presence of maternal antibodies, age-dependent maturation of the humoral immune response or co-infection with immunosuppressive viruses. MHC-defined differences between broiler lines did not influence the susceptibility to CAV when challenged by the oral route at four weeks of age (Joiner et al. 2005). However, differences in antibody response were noted between three MHC-defined white leghorn lines after CAV vaccination. The S13 (MHC: B¹³B¹³) line had a significantly lower seroconversion rate than the P2a (MHC: B¹⁹B¹⁹) and N2a (MHC: B²¹B²¹) lines (Cardona et al. 2000a). Earlier Hu (1992) had already suggested that S13 chickens were more susceptible to CAV than the N2a and P2a lines. Unfortunately, the S13 line is no longer available to investigate the basis for the observed differences.

Thus far, significant differences in pathogenicity between field isolates have not been reported. However, comparisons between strains based on publications from

different research groups are often impossible due to a lack of standardized procedures resulting in different virus doses and route of infection. Oral challenge often results in delayed virus replication and less severe lesions compared to intramuscular or intraabdominal injection (van Santen et al. 2004). Attenuation by cell culture passage or site-directed mutagenesis reduces the severity of lesions (von Bülow and Fuchs 1986; McKenna et al. 2003; Kaffashi et al. 2008). Antibody responses elicited by attenuated virus are similar to the responses invoked by pathogenic isolates after inoculation of five-week-old chickens (Todd et al. 1998).

9.4.2 Pathogenesis of infection in SPF chickens

The SPF flocks maintained by Cornell University were free of CAV (McNulty et al. 1989) until the mid 1990's. The flocks are maintained in a filtered-air, positive pressure house in a closed system with replacement flocks being hatched and reared in colony cages in separate rooms in the same building. These rooms have a common air source and the building is considered the unit. In 1996 the flocks became infected based on antibody testing (Cardona et al. 2000a). Since that time all birds in each flock of P2a and N2a lines are being tested for antibodies just prior to onset of egg production, between 34 and 40 weeks of age and at termination between 65 and 75 weeks of age. Seroconversion often occurs after the onset of lay and sometimes all birds within a flock become positive, but subsequent flocks may remain negative (Miller et al. 2001; Schat 2003; Schat unpublished data 2008). During the last 12 years clinical disease was never observed in the offspring of flocks that became antibody positive during the laying period (Schat unpublished data 2008), which is contrary to the experience with conventional flocks (see section 9.4.1.1).

Fig. 9.5

Fig. 9.6

These observations suggested a major difference between the pathogenesis in SPF flocks *versus* conventional flocks. The seminal finding by Cardona et al. (2000b) that viral DNA could be detected in the gonadal tissues of these SPF birds in the absence or presence of antibodies suggested the possible transfer of viral DNA to the newly hatched chicks without early viral replication leading to disease. Subsequently,

Miller et al. (2003) demonstrated the presence of viral DNA by nested PCR in blastodisk extracts and semen obtained from birds which were antibody negative between 432 and 494 days of age. Individual tissues from embryos obtained from antibody negative, blastodisk positive hens were examined for the presence of viral DNA at 18 days of embryonation. All embryos were positive but differences between tissues from individual embryos were noted indicating that the viral DNA was transmitted through the germ-line (Fig. 9.5). The low levels of viral DNA in the blastodisk and the presence in most tissues at 18 days of embryonation suggested the multiplication of DNA. Embryos were examined between zero and 12 days of incubation for the presence of viral DNA and transcripts by qPCR and qRT-PCR. Viral transcripts were detected between four and seven and between 10 and 12 days of embryonation (Fig 9.6), while the number of DNA copies remained very low (data not shown) (Miller, Stucker and Schat unpublished data). It is possible that virus is transferred from the blastoderm to the embryo through hematopoietic cells, because the first transient wave of hematopoietic precursors is derived from the yolk sac (Fellah et al. 2008). The intricate control of CAV transcription suggests a role for steroid hormones (see section 9.3.3.2). Gonadal development occurs around the time of the first wave of viral transcripts and it is hypothesized that estrogen is involved in the activation of viral transcription between four and seven days of embryonation. The second wave may be explained if hormone receptors develop during that period in for example the first wave of T lymphocytes, which start to appear around embryonation day 6 but may not be mature until shortly before the second wave of T lymphocytes appear between 10 and 12 days (Fellah et al. 2008).

The combination of strict control of viral transcription, the detection of viral DNA in gonads of adult hens in the absence of antibodies and the presence of minute amounts of viral DNA in blastodisks and embryonal tissues led Miller and Schat (2004) to propose that the interaction between CAV and the host is an example of the ultimate host-parasite relationship. In this model, latent virus or viral DNA may be passed from generation to generation without showing evidence of infection and occasionally may turn on virus replication resulting in seroconversion. Miller et al. (2003) tested this model by raising 3 generations in isolators starting with fertile eggs from antibody-negative hens. Each generation was tested frequently by PCR and

positive chicks were removed. Transmission may be intermittent: 1/9 chicks was positive from a breeding pair that had been selected for the production of the next generation based on the absence of viral DNA in 25 embryos.

Limited evidence suggests that vertical transmission of virus or viral DNA from VN antibody-positive hens to embryos may also occur in commercial breeder flocks (Brentano et al. 2005). It is currently not known if this is important for commercial flocks. It is important to note that vaccination of pullet breeders will not prevent this type of vertical transmission (Miller et al. 2003).

9.4.3 Consequences for immune responses

In section 4.1.1 it was already mentioned that infection of susceptible chickens causes a decrease in innate immune responses to bacterial infections. In addition to the decreased function of thrombocytes and heterophils, macrophage functions are also impaired. The number of macrophages in the spleen can be sharply decreased around 14 days pi (Cloud et al. 1992). Macrophages obtained from chickens infected by intramuscular injection at one day of age or by the oral route at 21 days of age had decreased expression of the FcR receptor, decreased phagocytic and bactericidal activity and decreased interleukin (IL)-1 production as measured by bioassay (McConnell et al. 1993a; 1993b). Unfortunately, these assays were performed at a time when few reagents were available to characterize the cells by macrophage-specific surface markers or by measuring IL-1 β by qRT-PCR. Markowski-Grimsrud and Schat (2003) used qRT-PCR assays on whole spleen extracts and were unable to demonstrate decreased or increased transcription rates for IL-1 β and IL-2 at 7 and 14 days pi, when chickens were infected between 9 and 45 days of age. They found a significant increase in interferon (IFN)- γ transcription at 7 days pi of 30-day-old birds, but the increase was also present in maternal antibody-positive birds, which did not replicate CAV. Adair et al. (1991) also reported an increase in IFN activity at 8 days pi using a bioassay, but a decrease at later sampling times.

Impairment of CTL activity is another important consequence of infection. Several groups noted a decrease in CD8⁺ cells often with an increase in ratio of CD4⁺ over CD8⁺ cells (Cloud et al. 1992; Bounous et al. 1995). As mentioned before a

marked decrease in CTL activity has been reported (Markowski-Grimsrud and Schat 2003), but NK cell activity was not affected (Markowski-Grimsrud and Schat 2001).

9.4.4 Interactions with other pathogens

Infection with immunosuppressive viruses can increase the severity of other infections or decrease the efficacy of vaccination against other pathogens. CAV is no exception in this respect and there are many examples in which CAV infection increased the severity of other diseases in experimental infections or under field conditions. Extensive recent reviews are available for detailed information on CAV-induced enhancement of other diseases (Miller and Schat 2004; Schat and van Santen, 2008) and only a few examples will be discussed. Toro et al. (2006b) showed a clear link between CAV infection and infectious bronchitis (IB), a coronavirus-induced respiratory disease in chickens, in IB-vaccinated chickens. This may seem puzzling because IB virus (IBV) vaccines protect mostly through VN antibodies, but IBV variants appear frequently in the field and vaccine-induced protection becomes suboptimal. In that case IBV-specific memory CTL become crucial for protection (Pei et al. 2003), which may be impaired by CAV.

MDV is another example of interaction of a second pathogen with CAV. In this case the interactions are more complicated than the interaction between CAV and IBV. Jakowski et al. (1970) noticed that birds developing anemia after challenge with MDV had a decreased incidence of MD lesions. It was subsequently shown that these birds were infected with MDV and CAV (see section 9.2). Von Bülow et al. (1983; 1986) experimentally infected one-day-old chicks with CAV, CAV and MDV or MDV. The dual challenge resulted in an increased degree of anemia and increased mortality but there was no clear impact on the MD incidence. These authors also confirmed that infections with IBDV or REV increased the CAV-induced anemia and mortality and suggested that these increases were caused by immunosuppression induced by MDV, IBDV and REV. The mechanisms of MDV-, IBDV-, and REV-induced immunosuppression are complex and have recently been reviewed (Schat and Skinner 2008). CAV can also influence MDV replication when both viruses are inoculated at the same time. For example, co-infection of CAV with the very virulent MDV strain RB-1B resulted in increased mortality and shorter mean time to death

compared to chickens inoculated with RB-1B alone. Interestingly chickens inoculated with the combination died without tumors in contrast to the birds receiving RB-1B alone. Co-infection caused an increase in transcription levels for the immediate early infected cell protein (ICP)4 of MDV in the spleen at 4, 7 and 13 days pi. If the very virulent + MDV isolate 584A was used instead of RB-1B this effect was only seen at 1 day pi. The effect of CAV on 584A-induced tumors was also less striking than with RB-1B, most likely because 584A caused a severe depletion of thymocytes and high levels of early mortality unrelated to tumor formation (Miles et al. 2001). In addition to the virulence of MDV, the challenge dose of MDV may also play a role in the interaction with CAV. Jeurissen and de Boer (1993) found that CAV significantly enhanced MD-associated lymphoproliferation in nerves and visceral organs using a relatively low dose of MDV, while coinfection with a 20-fold higher dose of MDV resulted in a decrease in lymphoproliferation compared to the high dose of MDV alone.

A more important interaction between CAV and MDV is the decrease in vaccinal immunity to MD as a consequence of CAV infection. Field observations have suggested that this may be an explanation for MD breaks as late as 24 weeks of age (Fehler and Winter 2001). Experimental data support the CAV-induced decrease in vaccinal immunity. Chickens vaccinated with HVT at one day of age and infected with CAV at 4 days of age and challenged with MDV were significantly less well protected than chickens that did not receive CAV. However when the MDV challenge was given at 18 days of age there was no negative effect by CAV (Yuasa and Imai 1988). Similar results were reported by Otaki et al. (1988). CAV replication decreases protective immunity by elimination of specific CTL (see section 9.4.3), which can occur when CAV infects young maternal antibody-negative birds or older antibody-negative birds or when CAV is reactivated during the development of sexual maturity and subsequent egg production. The absence of an effect when chickens are challenged with MDV at 18 days of age suggests that some CTL have escaped the CAV infection and can generate a memory response in the absence of active CAV replication.

4.5 Control of infection

Vaccines against CAV have been developed using various attenuated strains (reviewed by Schat and van Santen 2008) but the need for vaccination is not universally accepted (Smith 2006). Vaccines are considered to be expensive and the fact that immunosuppression may be the only consequence of infection complicates the diagnosis of CAV as an important part of disease syndromes. There are also some technical and practical problems affecting the widespread use of vaccines. The first problem is the relative instability of the virus and relative low levels of virus titers in cell cultures. The second problem is the testing of vaccines for efficacy, which requires vaccination followed by challenge demonstrating significant protection. Challenge is typically performed one to three weeks post vaccination, but induction of a high level of symptoms requires challenge before seven days of age. Markowski-Grimsrud et al. (2002) developed a qPCR assay differentiating between two strains so that one can determine if the vaccine virus protects against replication of the challenge virus. A practical problem is the timing of vaccination in young birds positive for maternal antibodies especially if a live vaccine is used. Similar problems have been noted with vaccinations for IBDV (Etteradossi and Saif 2008). Perhaps the use of antigen-antibody complex vaccines can overcome the problems with maternal antibodies as has been shown for this type of IBDV vaccines (Haddad et al. 1997).

5 Lessons for Anellovirus infections

The TTV and TTMV group shares certain similarities with CAV such as coding for proteins with DSP and apoptotic activity, which are discussed in more detail in chapter 6. The other similarity is the ubiquitous nature of TTV and TTMV infections in humans and of CAV in chickens. One of the key problems faced by the researchers studying the importance of TTV and TTMV is this high prevalence of these viruses in the human population without a clear link to one or more diseases. This situation is fairly similar to CAV, which also has a high prevalence and is not linked to a clinical disease syndrome unless the infection occurs in newly hatched, antibody-negative chicks or chickens with a severely impaired antibody response. The major impact that

CAV has is the subclinical infection aggravating other diseases by causing immunosuppression. This may be very similar in TTV and TTMV infections. Vertical mother to infant transfer of TTV has been reported (e.g., Xin et al. 2004), while others found no evidence for transmission through maternal blood. Instead they associated an increase in infection levels in infants with environmental infections (Ohto et al. 2002). The problem is that currently no methods have been widely used to determine if maternal antibodies are present to prevent virus replication either after vertical transmission or after early environmental infection. It may be of interest to determine in these studies if and when virus replicates using qRT-PCR as was done in the studies of CAV-induced immunosuppression by Markowski-Grimsrud and Schat (2003).

Many research teams have linked temporal increases in TTV load in HIV infected patients to cycles of HIV replication, which raises the question of what initiates the increase. Does an increase in HIV cause immunosuppression followed by an increase in TTV; is it the other way around or is it a vicious circle in which each episode of HIV or TTV replication enhances the replication of the other pathogen? TTV DNA titers were significantly higher in patients with AIDS, high HIV loads or low CD4⁺ T cell counts (Shibayama et al. 2001; Touinssi et al. 2001; Thom and Petrik 2007). The fact that TTV replicates in stimulated T cells (Mariscal et al. 2002) and bone marrow cells (Okamoto et al. 2000) certainly could lead to immunosuppression and thus lead to increased replication of HIV. This is similar to the replication of CAV in bone marrow cells and dividing T lymphocytes followed by enhanced replication of MDV. Girard et al. (2007) suggested that TTV may be a co-factor in the development of Kaposi's sarcoma, which supports the hypothesis that subclinical immunosuppression caused by TTV infection may increase the development of other diseases. In parallel with increased replication of CAV in immunosuppressive treatments in chickens, TTV replication increases when transplant patients receive immunosuppressive treatments (Moen et al. 2003). It will be of interest to determine if immunosuppressive treatment affecting principally the humoral or the cell-mediated responses are causing the increase in virus load and in which cell the virus replicates under those conditions.

6 Conclusions

CAV is an important pathogen causing economic losses by increasing the clinical severity of diseases caused by other pathogens, mostly as a consequence of subclinical infection. This virus with a small genome has a very complex mechanism to maintain itself in low density or low stress populations and the elucidation of the control of latency may unravel some fundamental aspects of gene control. The induction of apoptosis by VP3 is a crucial aspect of viral replication and subsequent pathology, but the mechanism of apoptosis in nontransformed chicken cells is poorly understood and needs additional study. Identification of the substrate(s) for the DSP activity of VP2 will be important for the further development of attenuated strains and perhaps vaccines. In general a better understanding of the biology of CAV may lead to insight on the role of TTV in human health and disease.

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Fig 9.1 Phylogenic analysis of the predicted amino acid sequences of VP1 of 26 new CAV isolates from Guangdong and Huan provinces, China and 47 VP1 sequences available on GenBank. The Nobilis® P4 vaccine strain is also included. Numbers at nodes correspond to bootstrap values >49. Chinese strains described by Ducatez et al. (2008) are marked with a closed triangle, whereas other

Chinese sequences from Genbank have open triangles. From Ducatez et al. 2008 Avian Dis 52:68-73, with permission from the authors.

Fig. 9.2 Schematic representation of the CAV transcripts and their candidate ORFs. (a) Six ORFs of the CAV genome. Short and long vertical lines indicate ATG start and stop codons, respectively. Putative ORFs encoded by the transcripts are indicated by arrows. ORFs evident in the 2.0, 1.3, 1.2 and 0.8 kb transcripts are shown in panels (b)-(e), respectively. Dotted lines indicate introns or deleted regions. Black boxes represent putative coding regions. Nucleotide numbers indicate the positions of ORF junctions or boundaries of ORFs. From Kamada et al. 2006 J Gen Virol 87:2227-2233, with permission from the Society for General Microbiology.

Fig. 9.3 (a) Schematic diagram of the CAV promoter region with the areas used for the short and long promoter vectors indicated by the bars under the diagram. The CAV promoter 21 bp direct repeat sequences are TGTACAGGGGGGT**ACGTC**ACCCGTACAGGGGGGT**ACGTC**ACA, with the estrogen response element (ERE)-like sequences within the 21 bp DR in boldface underlined type. The ERE-like sequences arranged as direct repeats with a 15 bp separation are indicated in the diagram as boxes labeled DR-15. Known or putative transcription factor binding sites are indicated as circles for Sp1, squares for ERE-like and triangles for NFY. (b) EGFP expression compared in LMH and LMH-2A cells transfected with pEGFP-SE and pEGFP-LE, with or without 10^{-7} M 17β -estradiol (E2). Results are the average of 3 or more independent experiments, each with triplicate wells. EGFP expression was measured by fluorometer and normalized as a percentage of the positive-control CMV promoter-driven vector. (c and d) qRT-PCR analysis of EGFP mRNA in LMH-2A (C) and DF-1 (D) cells transfected with positive control CMV promoter plasmid pEGFP-N1, or the CAV promoter constructs pEGFP-S5 or pEGFP-L5. Results are from one experiment with 5 separate transfected wells. The results are reported as mRNA copy numbers and are normalized for transfection efficiency by dividing by DNA copies of the expression plasmid. RNA and DNA EGFP copies were normalized to the cellular genes GAPDH and iNOS, respectively. Error bars in (b), (c) and (d) indicate standard deviations, columns with different letters are significantly different with a P value of <0.05 by two-tailed t test analysis. From Miller et al. 2005 J Virol 79:2859-2868, with permission from the American Society of Microbiology.

Fig. 9.4 Thymus of a chicken inoculated at 1 day of age with 3000 chicken infective doses 50% of the CIA-1 strain of CAV and harvested at 14 days post infection. $CD4^{+}$ cells are stained red and $CIA-1^{+}$ cells are stained blue. Notice the low numbers of $CD4^{+}$ cells and the relatively few $CIA-1^{+}$ cells. From Hu 1992 MS thesis Cornell University, with permission from the author.

Fig. 9.5 Distribution of CAV in embryos from five hens, each designated by a three-digit number. These hens were antibody negative but had PCR-positive blastodisks. The hens were inseminated with semen from one male that was PCR positive on 2 of 4 semen samples tested and was also positive on

postmortem samples of the testicle and vas deferens. Hen #647 became seropositive during the period of egg collection. In total 28 embryos were examined. Columns that have the same lowercase superscript are significantly different at $P < 0.05$ by chi-square analysis. From Miller et al. 2003 Avian Dis 47:662-671 with permission from the American Association of Avian Pathologists.

Fig. 9.6 Determination of CAV transcripts in embryos using real-time quantitative RT-PCR. Total RNA was extracted from 10 fertile blastoderm (B) samples and 10 embryos/sampling day at embryonation day (ED) 1 and 3-9 in Expt. 1 (Panel a), and from 10 blastoderm (B) samples and 10 embryos/sampling day at ED 1-12 in Expt. 2 (Panel b). RNA copy numbers were normalized for equal loading using the cellular housekeeping gene, GAPDH. The results are reported as log 2 of the average copy number and error bars indicate the standard error of the mean. Numbers above the bars indicate the number positive/10 samples. Sampling days without a number had no positive samples. Unpublished data from M.M. Miller, K.A. Stucker and K.A. Schat, with permission from Miller and Stucker.

Table 9.1 Comparison of predicted amino acid sequence for VP1 from pathogenic Cux-1 and clone AH-C364 and the attenuated clones P310 clone 34 and AHC140

Isolate	Amino acid changes and position																	
	29	75	89	125	140	141	144	251	254	265	287	321	370	376	394	413	444	447
Cux-1 ^a	K	V	T	I	S	Q	D	Q	E	N	A	R	S	L	Q	A	Y	T
P310 C 34 ^b	R	I	A	L	S	L	E	L	G	T	A	A	S	L	Q	A	Y	T
AH-C140 ^c	R	V	T	L	A	Q	E	R	E	T	S	A	G	I	H	S	Y	S
AH-C364 ^c	R	V	T	L	A	E/Q ^d	E	R	E	T	S	A	G	I	Q	S	D ^e	S

^aMeehan et al. 1992, ^bScott et al. 2001, ^cYamaguchi et al. 2001, ^dAn additional pathogenic clone had Q at this position,

^eAn additional pathogenic clone had Y at this postion.

Table 9.2 Biological effects of mutations in the DSP motif (mutants 1-4) and in the sequence important for tertiary structure (mutants 5-11).

Mutant		Effect of mutation in:						
No	Change in amino acid	MDCC-MSB1 ^a			Embryos ^b		Bird inoculation ^c	
		Virus rescue	Cpe in MSB1	MHC-I decrease	Apop-tosis	Lesion score	Relative thymus weight	Thymic cortex
	Wild-type (CAU269/7)	+	+++	decrease	yes	20	decreased	decreased
1	C87R ^d	+	+	... ^e	...	9	...	...
2	R101G	+	+	normal	...	11	...	...
3	K102D	- ^a	+	...	...	...	...	...
4	H103Y	+	+	normal	...	11	...	...
5	R129G	+	++		no	10	...	...
6	Q131P	+	++	intermediate	no	12	normal	...
7	R/K/K150/151/152G/A/A	+	++	...	no	12	decreased	normal
8	D/E161/162/G/G	+	+	...	...	11	...	...
9	L163P	+	++	...	no	15	...	...
10	D169G	+	+	...	...	15	...	...
11	E186G	+	++	...	...	9	decreased	normal

12	S77N ^f		decreased	normal ^g
	Negative control medium	7	normal	normal

^aPeters et al. 2006, ^bPeters et al. 2007, ^cKaffashi et al. 2008, ^dC86R in Peters et al. 2006, ^e ... no information provided or not done, ^fderived from R129G, ^gnormal compared to one medium control and decreased compared to a second medium control group.

Abbreviations:

ALV	avian leukosis virus
CAV	chicken (infectious) anemia virus
COUP-TF1	chicken ovalbumin upstream promoter transcription factor 1
CTL	cytotoxic T lymphocytes
DR	direct repeat(s)
ds	double-stranded
EGFP	enhanced green fluorescent protein
ERE	estrogen response element
HRE	hormone response elements
HVT	herpesvirus of turkeys
IBD(V)	infectious bursal disease (virus)
IB(V)	infectious bronchitis (virus)
MAb	monoclonal antibodies
MDCC	Marek's disease chicken cell line
MDV	Marek's disease virus
MHC	major histocompatibility complex
NK	natural killer
nt	nucleotides
p	passage
pi	post infection
PCR	polymerase chain reaction
PCV	porcine circovirus
q(RT-)PCR	quantitative (reverse transcription-) PCR
RF	replicative form
REV	reticuloendotheliosis virus
SPF	specific-pathogen-free
TCR	T cell receptor
TTV	torque teno virus
TTMV	torque teno mini virus
VN	virus-neutralizing
VP	viral protein

